MP307 Practical 4 Continuous Population Models

Name: Cian Mac Liatháin ID: 11476078

In [76]:

```
%matplotlib notebook
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import odeint
```

Q1.(*) Verhulst Logistic Model.

Consider the logistic population model

$$\frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right)$$

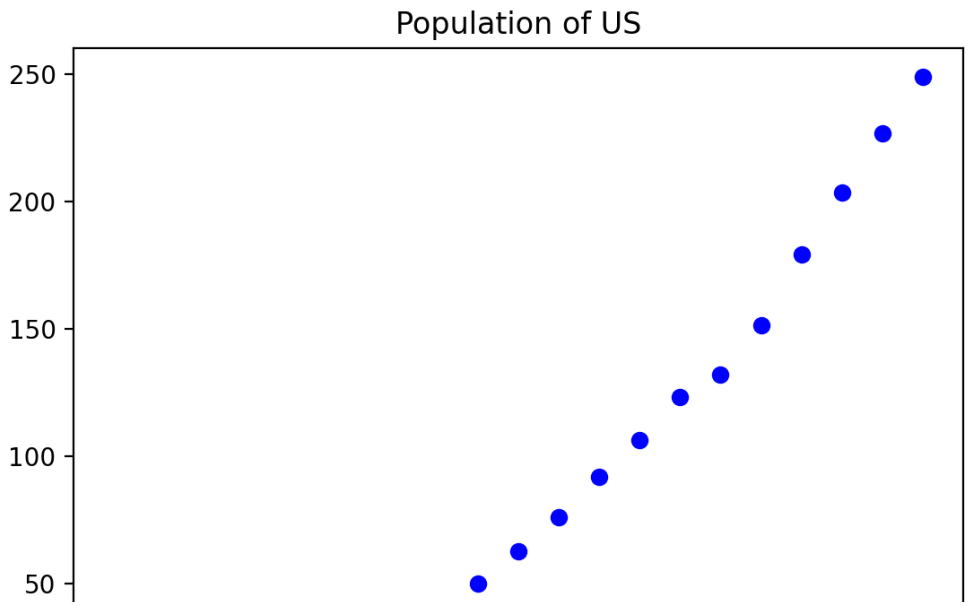
for $r, K > 0$ as a model of the US population from 1790 to 1990. The data for this are as given in the worksheet Q2. of `lab3_discrete_population_models.ipynb` as follows:

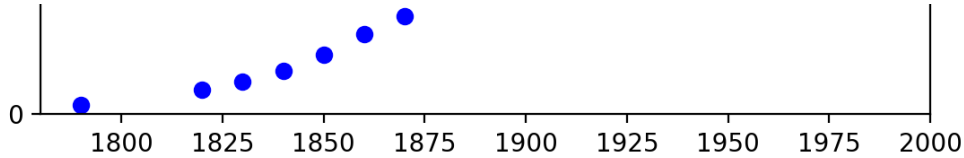
In [77]:

```
Year = [1790, 1820, 1830, 1840, 1850, 1860, 1870, 1880, 1890, 1900, 1910, 
        1940, 1950, 1960, 1970, 1980, 1990]
Pop= [3.79, 9.6, 12.9, 17.1, 23.2, 31.4, 38.6, 50.2, 62.9, 76.0, 92.0, 106
      132.0, 151.3, 179.3, 203.3, 226.54, 248.7]
```

In [78]:

```
# plot of data
plt.figure(1)
plt.plot(Year,Pop,'bo')
plt.axis([1780,2000,0,260])
plt.title('Population of US')
plt.show()
```





The Logistic differential equation has solution

$$P(t) = \frac{K}{1 + \left(\frac{K}{P_0} - 1\right) e^{-rt}}$$

In [79]:

```
#Logistic Model

# Python function for Logistic solution where t=0 is 1790
def P(t):
    P_0=3.79
    r=0.3 #  in inverse decades
    K=300.0
    ans=K/(1.+(K/P_0-1.)*np.exp(-r*t))
    return ans
```

In [80]:

```
P(0)
```

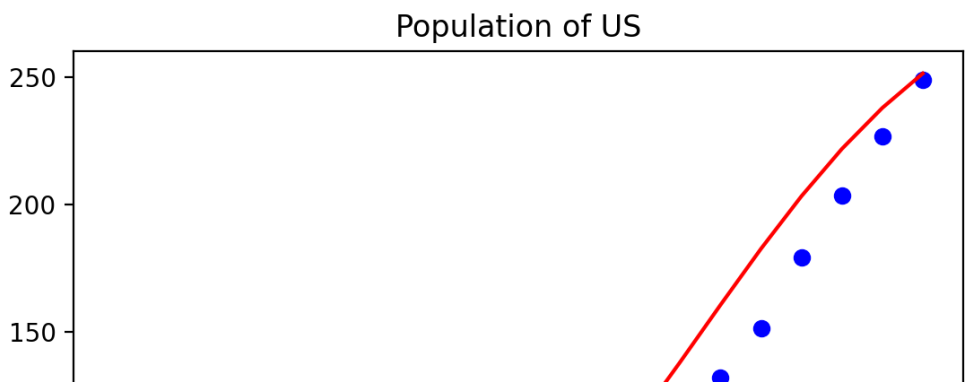
Out[80]: 3.79

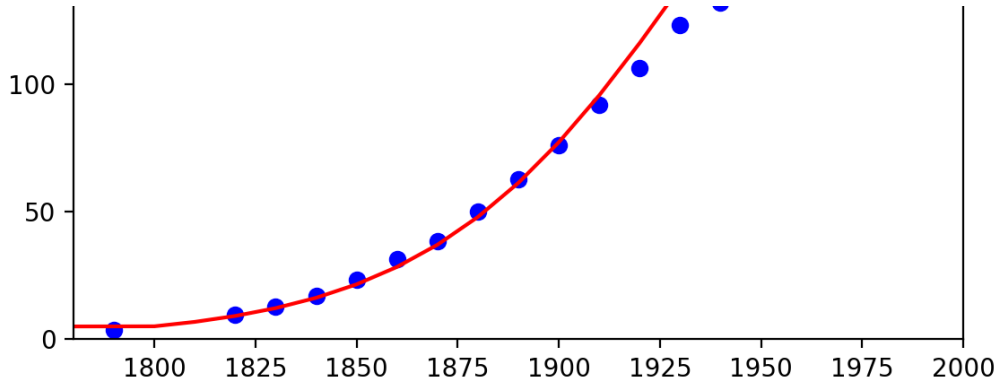
In [81]:

```
# plot Verhulst pop PV from 1790 to 1990 with 10 year intervals
PV=[0]*21 # list for Verhulst P initialized with 20 zeros
YV=[0]*21 # list for years initialized with 20 zeros

# choose 10 year intervals
for n in range(1,21):
    YV[n]=1790+10*n
    PV[n]=P(n)
```

In [82]:

```
plt.figure(2)
plt.plot(Year,Pop,'bo')
plt.plot(YV,PV,'r-')
plt.axis([1780,2000,0,260])
plt.title('Population of US')
plt.show()
```





In [ ]:

Q.2(*) Competitive Species Model

Consider two species with population sizes P_1, P_2 with growth rates r_1, r_2 and limiting population sizes of K_1, K_2 where

$$\frac{dP_1}{dt} = r_1 P_1 \left(1 - \frac{P_1 + P_2}{K_1}\right),$$
$$\frac{dP_2}{dt} = r_2 P_2 \left(1 - \frac{P_1 + P_2}{K_2}\right).$$

Analyse the behaviour of P_1, P_2 in the following cases, by plotting P_1, P_2 vs t and P_1 vs P_2:

(a). $r_1 = 1/10$, $r_2 = 1/10$, $K_1 = 100$, $K_2 = 50$ with $P_1(0) = 10$ and $P_2(0) = 15$.

(b). $r_1 = 1/10$, $r_2 = 1/10$, $K_1 = 100$, $K_2 = 50$ with $P_1(0) = 130$ and $P_2(0) = 200$.

(c). $r_1 = 1/10$, $r_2 = 1/100$, $K_1 = 40$, $K_2 = 50$ with $P_1(0) = 130$ and $P_2(0) = 20$.

(d). $r_1 = 1/10$, $r_2 = 1/100$, $K_1 = 50$, $K_2 = 60$ with $P_1(0) = 15$ and $P_2(0) = 10$.

In [112…

```
#(a)
# parameter values
r_1, r_2 = 1/10, 1/10

K_1, K_2 = 100, 50

P1_0, P2_0 = 10, 15 # initial populations
```

Define a function `dP_dt(P, t)` of a Python list `P=[P1,P2]` describing dP1/dt and dP2/dt.

Note: recall that indexing in Python goes 0,1,... so that Pop1 is stored in `P[0]` and Pop2 is stored in `P[1]`

In [113…

```
def dP_dt(P, t): # P=[P1,P2] list
    return [r_1*P[0]*(1 - (P[0]+P[1])/K_1), r_2*P[1]*(1 - (P[0]+P[1])/K_2)
```

Use `odeint` Python function to integrate the pair of odes subject to the initial conditions for time from `tmin` to `tmax` .

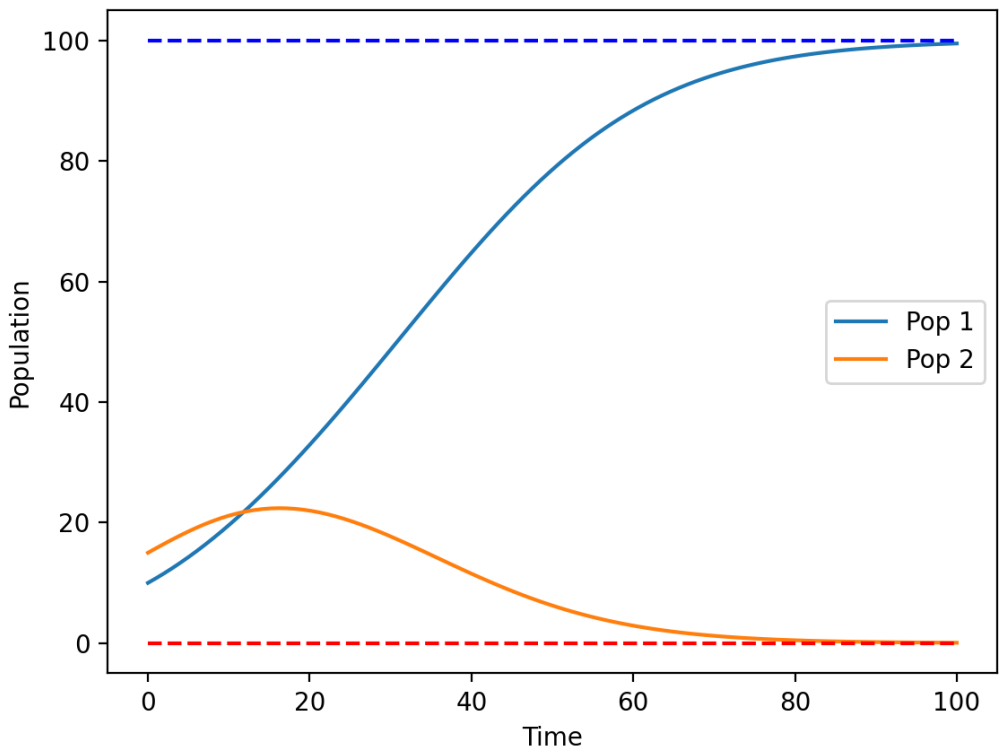
In [114…

```
tmin=0
tmax=100.

ts = np.linspace(tmin, tmax, 100) # 100 equally spaced t values from tmin t
P0 = [P1_0, P2_0] #initial conditions
Ps = odeint(dP_dt, P0, ts)
P1 = Ps[:,0] # values of P1 on ts values
P2 = Ps[:,1] # values of P2 on ts values
```

Plot P_1 and P_2 vs t and asymptotic lines.

In [115…

```
plt.figure(3)
plt.plot(ts, P1, "-", label="Pop 1")
plt.plot(ts, P2, "-", label="Pop 2")
plt.xlabel("Time")
plt.ylabel("Population")
plt.legend();
```



Notice that as $t \to \infty$ then $(P_1, P_2) \to (K_1, 0)$ for $K_1 > K_2$ and $(P_1, P_2) \to (0, K_2)$ for $K_1 < K_2$.

We can add the asymptotic lines to the above graph as follows

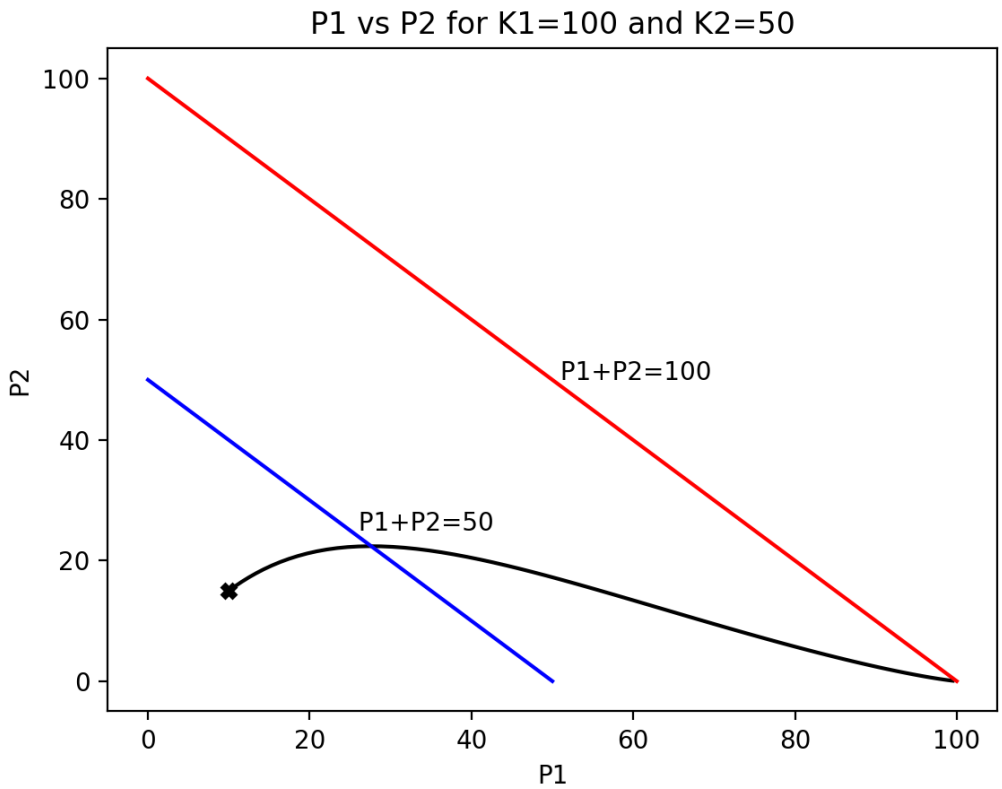
In [116…

```
P1_infty, P2_infty = K_1, 0
if K_2>K_1:
    P1_infty, P2_infty = 0,K_2
plt.plot([tmin,tmax],[P1_infty,P1_infty],"b--")
plt.plot([tmin,tmax],[P2_infty,P2_infty],"r--");
```

Plot P_1 vs P_2 with initial point marked with an x. We also plot the lines $P_1 + P_2 = K_1$ and $P_1 + P_2 = K_2$.

In [117…

```
plt.figure(4)

plt.plot(P1_0, P2_0,'kX') # mark initial point with x

plt.plot(P1, P2, 'k-')
plt.plot([0,K_1],[K_1,0],'r-')
plt.plot([0,K_2],[K_2,0],'b-')
plt.text(K_1/2, K_1/2,' P1+P2='+str(K_1))
plt.text(K_2/2, K_2/2,' P1+P2='+str(K_2))
plt.xlabel("P1")
plt.ylabel("P2")
plt.title("P1 vs P2 for K1="+str(K_1)+" and K2="+str(K_2));
```



In [120…

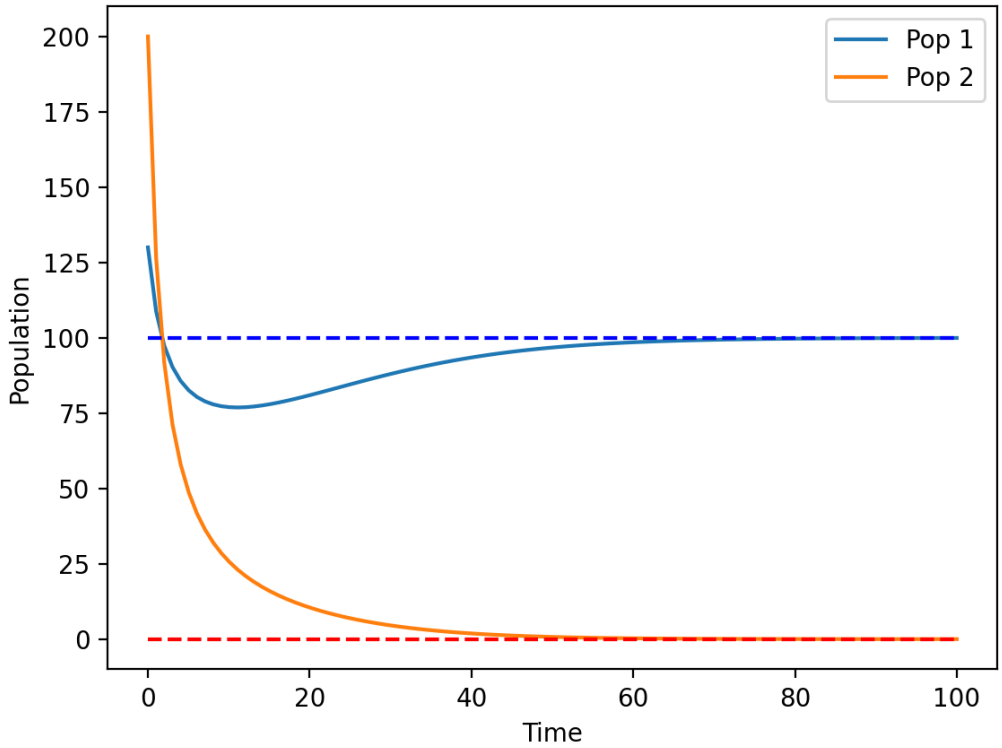
```
# (b)
# parameter values
r_1, r_2 = 1/10, 1/10

K_1, K_2 = 100, 50

P1_0, P2_0 = 130, 200 # initial populations
```

In [121…

```
tmin=0
tmax=100

ts = np.linspace(tmin, tmax, 100) # 100 equally spaced t values from tmin t
P0 = [P1_0, P2_0] #initial conditions
Ps = odeint(dP_dt, P0, ts)
P1 = Ps[:,0] # values of P1 on ts values
P2 = Ps[:,1] # values of P2 on ts values
```

In [122…

```
plt.figure(5)
plt.plot(ts, P1, "-", label="Pop 1")
plt.plot(ts, P2, "-", label="Pop 2")
plt.xlabel("Time")
plt.ylabel("Population")
plt.legend();
```

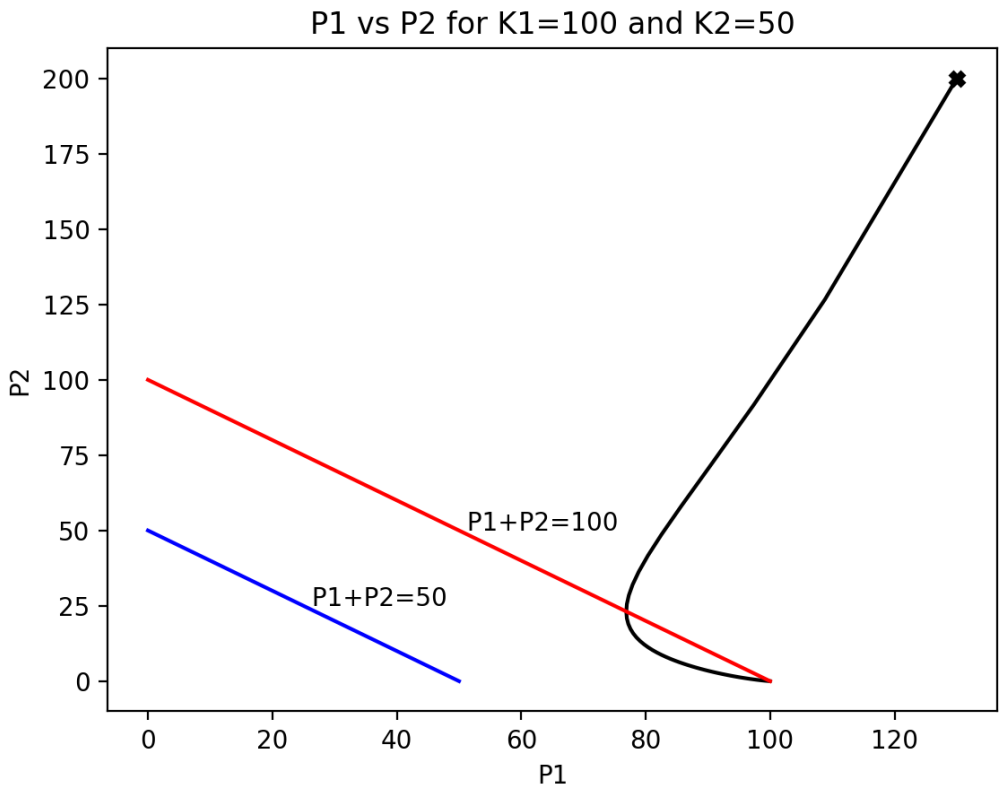


In [123…

```
P1_infty, P2_infty = K_1, 0
if K_2>K_1:
    P1_infty, P2_infty = 0,K_2
plt.plot([tmin,tmax],[P1_infty,P1_infty],"b--")
plt.plot([tmin,tmax],[P2_infty,P2_infty],"r--");
```

In [124…

```
plt.figure(6)

plt.plot(P1_0, P2_0,'kX') # mark initial point with x

plt.plot(P1, P2, 'k-')
plt.plot([0,K_1],[K_1,0],'r-')
plt.plot([0,K_2],[K_2,0],'b-')
plt.text(K_1/2, K_1/2,' P1+P2='+str(K_1))
plt.text(K_2/2, K_2/2,' P1+P2='+str(K_2))
plt.xlabel("P1")
plt.ylabel("P2")
plt.title("P1 vs P2 for K1="+str(K_1)+" and K2="+str(K_2));
```



In [125…

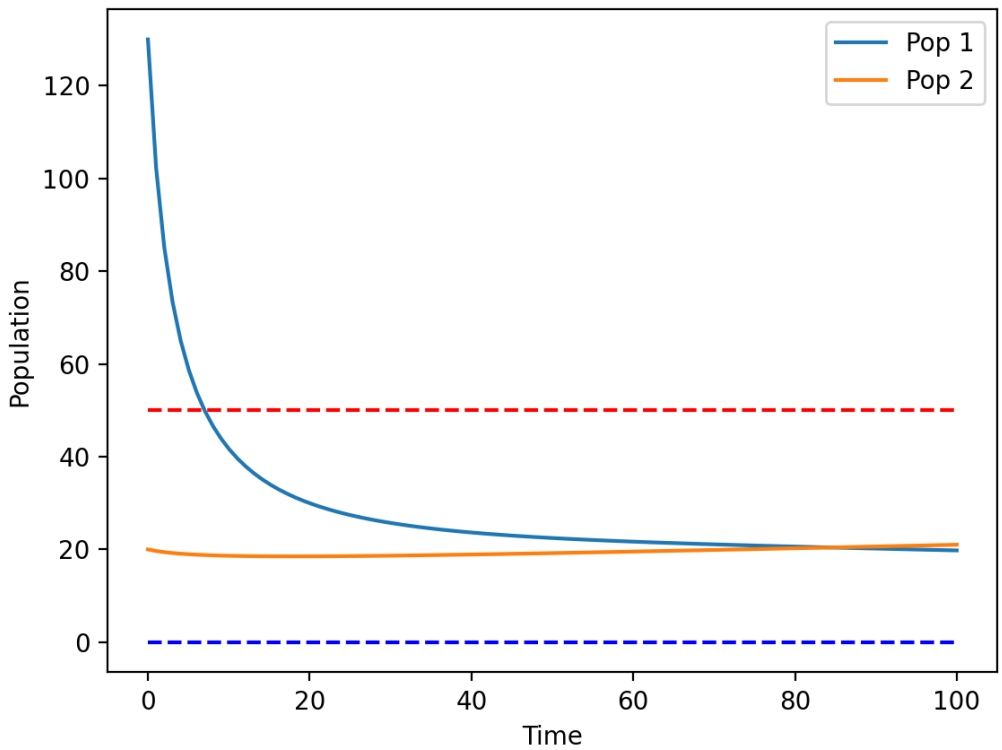
```
#(c)
# parameter values
r_1, r_2 = 1/10, 1/100

K_1, K_2 = 40, 50

P1_0, P2_0 = 130, 20 # initial populations
```

In [126…

```
P0 = [P1_0, P2_0] #initial conditions
Ps = odeint(dP_dt, P0, ts)
P1 = Ps[:,0] # values of P1 on ts values
P2 = Ps[:,1] # values of P2 on ts values
```

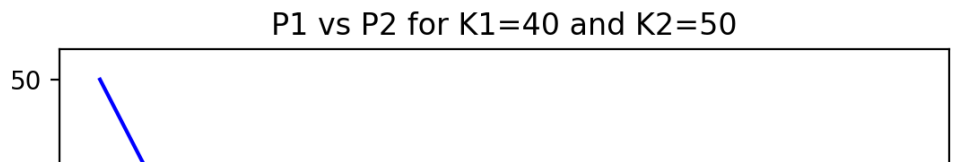
In [127…

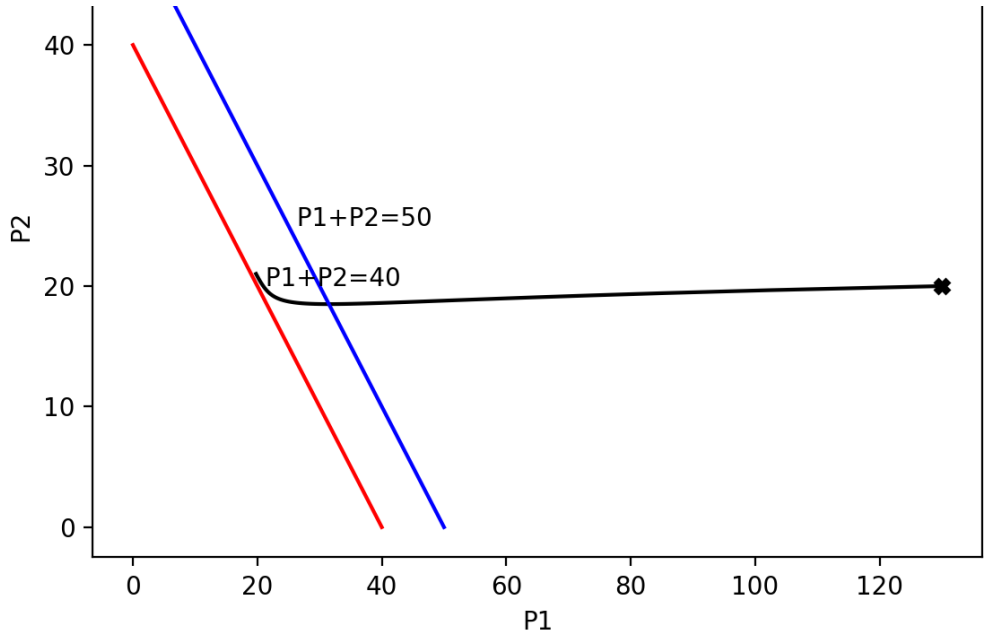
```
plt.figure(7)
plt.plot(ts, P1, "-", label="Pop 1")
plt.plot(ts, P2, "-", label="Pop 2")
plt.xlabel("Time")
plt.ylabel("Population")
plt.legend();
P1_infty, P2_infty = K_1, 0
if K_2>K_1:
    P1_infty, P2_infty = 0,K_2
plt.plot([tmin,tmax],[P1_infty,P1_infty],"b--")
plt.plot([tmin,tmax],[P2_infty,P2_infty],"r--");
```



In [128…

```
plt.figure(8)

plt.plot(P1_0, P2_0,'kX') # mark initial point with x

plt.plot(P1, P2, 'k-')
plt.plot([0,K_1],[K_1,0],'r-')
plt.plot([0,K_2],[K_2,0],'b-')
plt.text(K_1/2, K_1/2,' P1+P2='+str(K_1))
plt.text(K_2/2, K_2/2,' P1+P2='+str(K_2))
plt.xlabel("P1")
plt.ylabel("P2")
plt.title("P1 vs P2 for K1="+str(K_1)+" and K2="+str(K_2));
```





In [130…

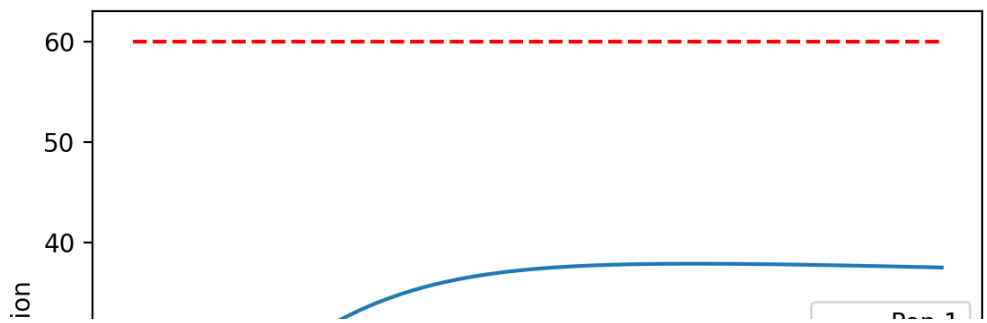
```
# (d)
# parameter values
r_1, r_2 = 1/10, 1/100

K_1, K_2 = 50, 60

P1_0, P2_0 = 15, 10 # initial populations
```

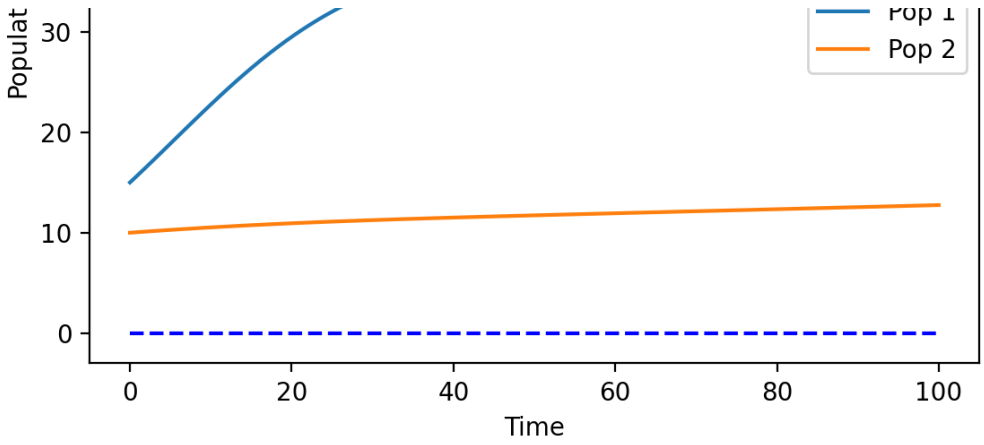
In [131…

```
P0 = [P1_0, P2_0] #initial conditions
Ps = odeint(dP_dt, P0, ts)
P1 = Ps[:,0] # values of P1 on ts values
P2 = Ps[:,1] # values of P2 on ts values
```

In [132…

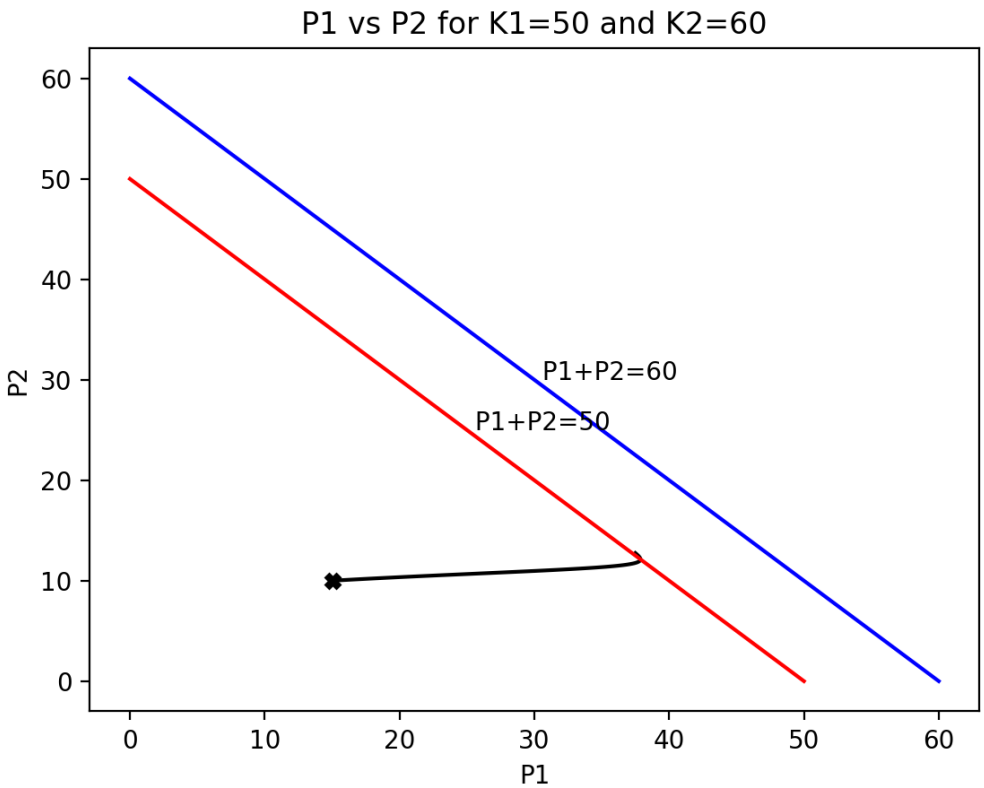
```
plt.figure(9)
plt.plot(ts, P1, "-", label="Pop 1")
plt.plot(ts, P2, "-", label="Pop 2")
plt.xlabel("Time")
plt.ylabel("Population")
plt.legend();
P1_infty, P2_infty = K_1, 0
if K_2>K_1:
    P1_infty, P2_infty = 0,K_2
plt.plot([tmin,tmax],[P1_infty,P1_infty],"b--")
plt.plot([tmin,tmax],[P2_infty,P2_infty],"r--");
```





In [133…

```
plt.figure(10)

plt.plot(P1_0, P2_0,'kX') # mark initial point with x

plt.plot(P1, P2, 'k-')
plt.plot([0,K_1],[K_1,0],'r-')
plt.plot([0,K_2],[K_2,0],'b-')
plt.text(K_1/2, K_1/2,' P1+P2='+str(K_1))
plt.text(K_2/2, K_2/2,' P1+P2='+str(K_2))
plt.xlabel("P1")
plt.ylabel("P2")
plt.title("P1 vs P2 for K1="+str(K_1)+" and K2="+str(K_2));
```



Q.3(*) Lotka-Volterra Predator/Prey System

Consider a prey species with population size x and a predator species with population size y where

$$\frac{dx}{dt} = x(a_1 - b_1 y),$$
$$\frac{dy}{dt} = y(-a_2 + b_2 x),$$

with $a_1, a_2, b_1, b_2 > 0$. Analyse the behaviour of the system for $a_1 = 3, a_2 = 5/2$ and $b_1 = 2, b_2 = 1$ by plotting x, y vs t, and x vs y in the following cases:

(a). $x(0) = 1$ and $y(0) = 1$.

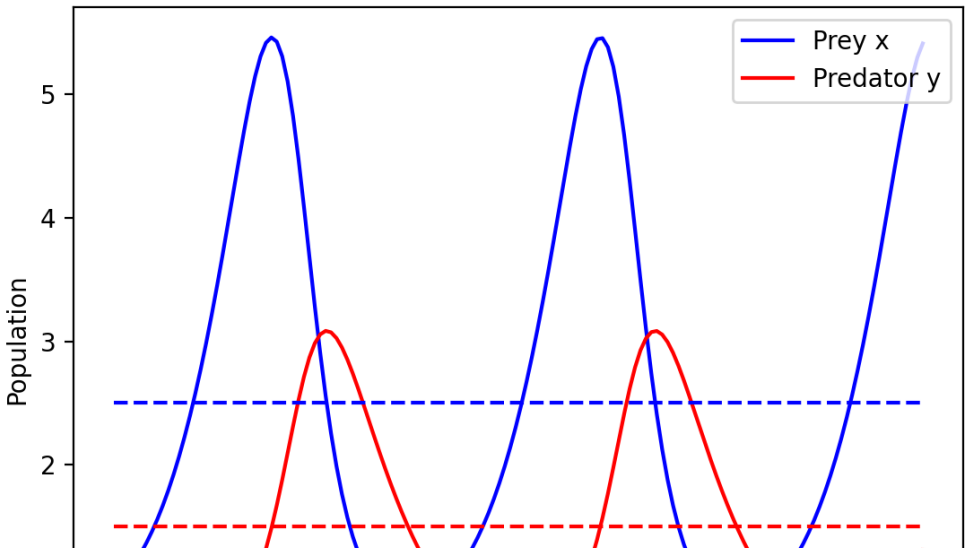
(b). $x(0) = 0.1 + 5/2$ and $y(0) = 0.1 + 3/2$. What behaviour do you observe?

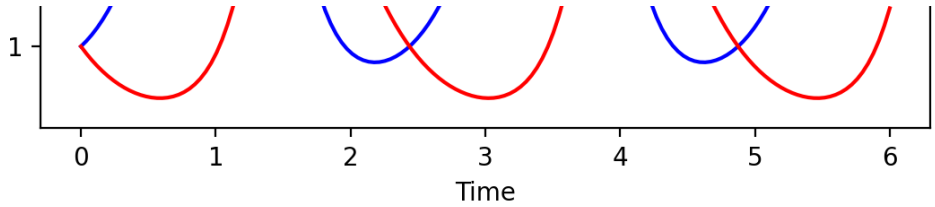
(c). $x(0) = 1$ and $y(0) = 5$.

Use similar coding to above

In [134…

```
#(a)
a_1, a_2, b_1, b_2 = 3, 5/2, 2, 1

x0, y0 = 1, 1 # initial populations

timerange=6.

def dP_dt(P, t): # P=[x,y] list
    return [P[0]*(a_1 - b_1*P[1]), P[1]*(-a_2 +b_2 *P[0])]

ts = np.linspace(0, timerange, 150)
P0 = [x0, y0] #initial conditions
Ps = odeint(dP_dt, P0, ts)
prey = Ps[:,0]
predators = Ps[:,1]

plt.figure(11)
plt.plot(ts, prey, "b-", label="Prey x")
plt.plot(ts, predators, "r-", label="Predator y")
plt.xlabel("Time")
plt.ylabel("Population")
plt.legend();
```





Equlibrium values $x_{eq} = a_2/b_2$ and $y_{eq} = a_1/b_1$. Notice that the maximum/minimum of x occurs when $y = y_{eq}$ and similarly, the maximum/minimum of y occurs when $x = x_{eq}$

In [135…

```
xeq , yeq =a_2/b_2, a_1/b_1
print("(xeq,yeq)=",(xeq,yeq))
```

```
(xeq,yeq)= (2.5, 1.5)
```

Look at values of x for $y = y_{eq}$ and of y for $x = x_{eq}$ in the last plot

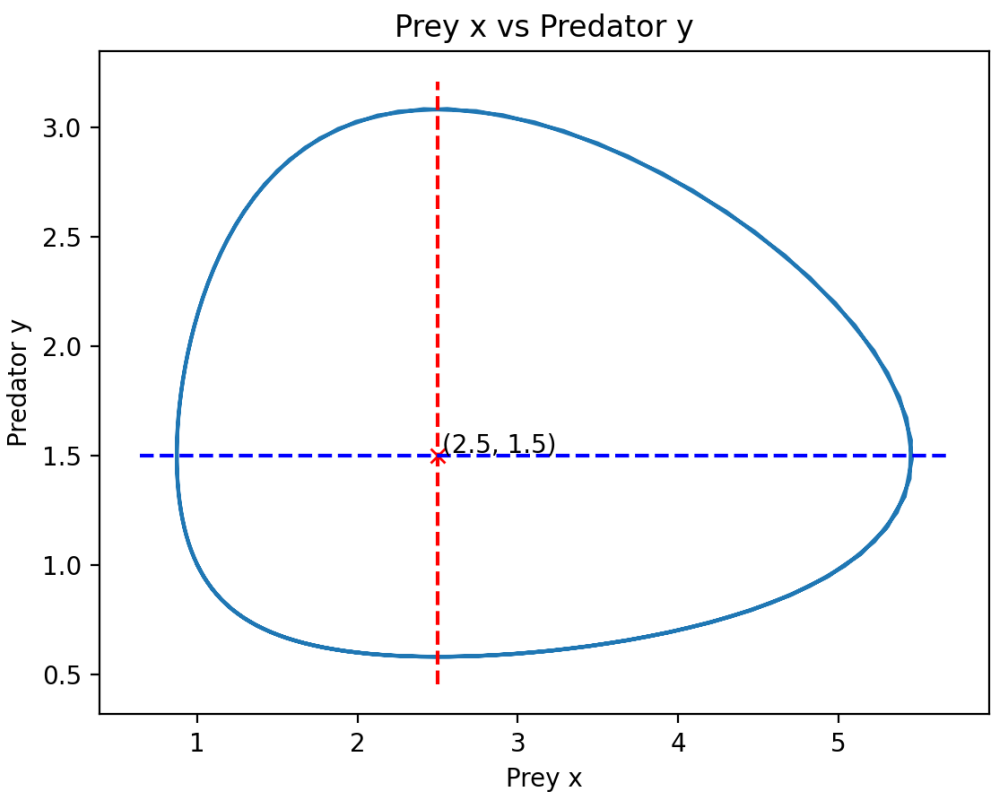
In [136…

```
plt.plot([0,timerange],[xeq,xeq],"b--")
plt.plot([0,timerange],[yeq,yeq],"r--");
```

Plot x vs y and show the equilibrium point. This plot shows that the system is periodic.

In [137…

```
plt.figure(12)
plt.plot(prey, predators, "-")
plt.plot([xeq],[yeq],'rx')
plt.text(xeq*(1.01),yeq*(1.01),str((xeq,yeq)))
plt.xlabel("Prey x")
plt.ylabel("Predator y")
plt.title("Prey x vs Predator y");
```



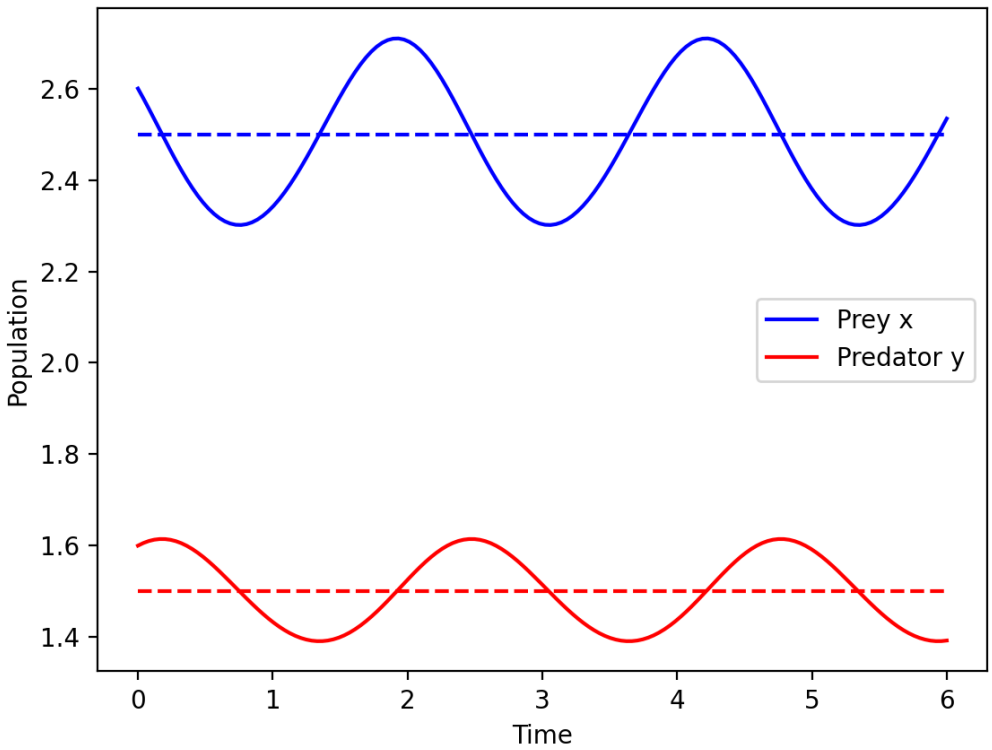
Add lines $x = x_{eq}$ and $y = y_{eq}$ to this plot

In [138…

```
miny=min(predators)
maxy=max(predators)
dely=maxy-miny
plt.plot([xeq,xeq],[miny-dely*0.05,maxy+dely*0.05],"r--")

minx=min(prey)
maxx=max(prey)
delx=maxx-minx
plt.plot([minx-delx*0.05,maxx+delx*0.05],[yeq,yeq],"b--");
```

In [139…

```
#(b)
a_1, a_2, b_1, b_2 = 3, 5/2, 2, 1

x0, y0 = 0.1+5/2, 0.1+3/2 # initial populations

timerange=6.

ts = np.linspace(0, timerange, 150)
P0 = [x0, y0] #initial conditions
Ps = odeint(dP_dt, P0, ts)
prey = Ps[:,0]
predators = Ps[:,1]

plt.figure(13)
plt.plot(ts, prey, "b-", label="Prey x")
plt.plot(ts, predators, "r-", label="Predator y")
plt.xlabel("Time")
plt.ylabel("Population")
plt.legend();
```

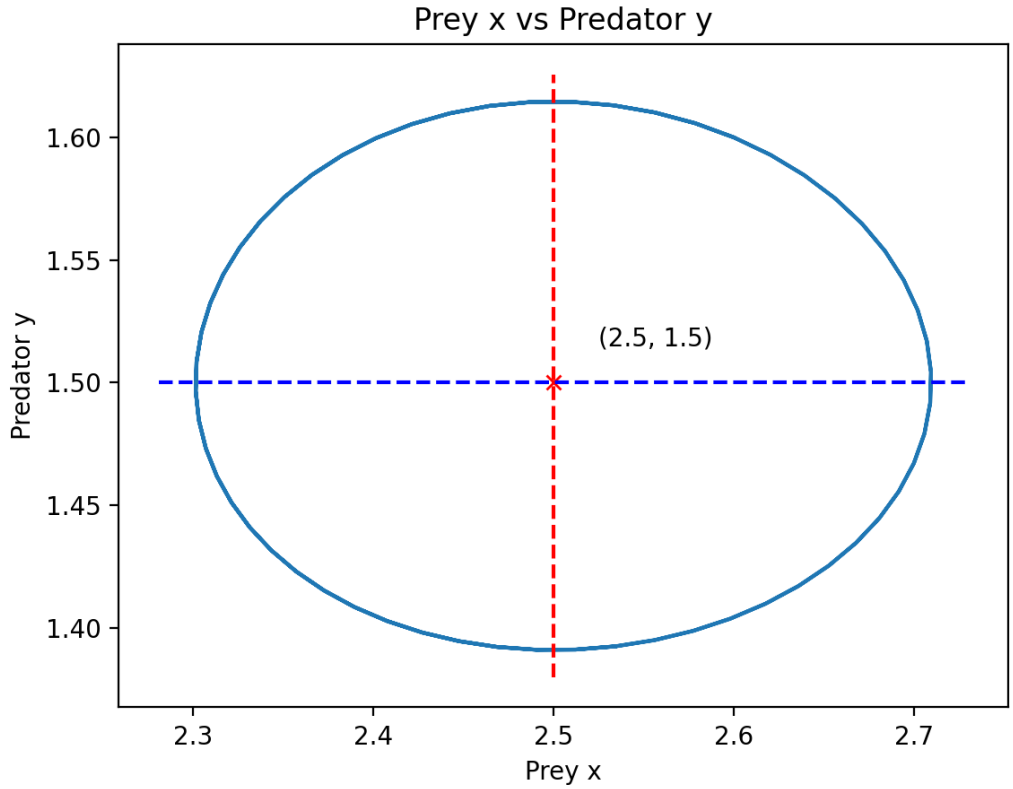


In [141…

```
plt.plot([0,timerange],[xeq,xeq],"b--")
plt.plot([0,timerange],[yeq,yeq],"r--");
```

In [142…

```
plt.figure(14)
plt.plot(prey, predators, "-")
plt.plot([xeq],[yeq],'rx')
plt.text(xeq*(1.01),yeq*(1.01),str((xeq,yeq)))
plt.xlabel("Prey x")
plt.ylabel("Predator y")
plt.title("Prey x vs Predator y");
```

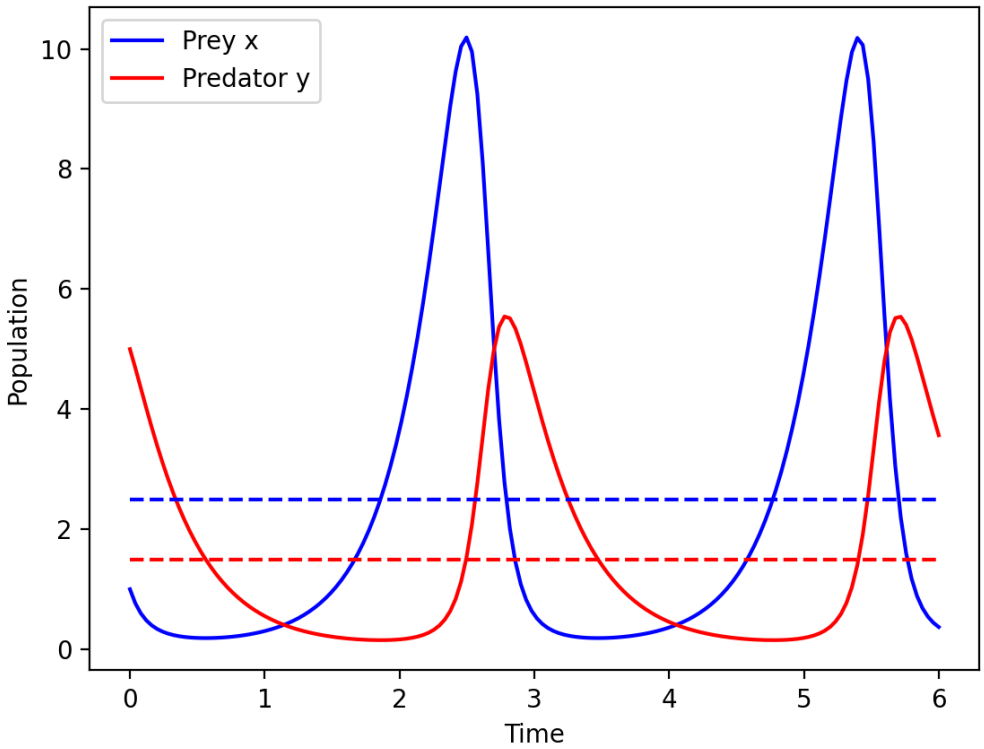


In [143…

```
miny=min(predators)
maxy=max(predators)
dely=maxy-miny
plt.plot([xeq,xeq],[miny-dely*0.05,maxy+dely*0.05],"r--")

minx=min(prey)
maxx=max(prey)
delx=maxx-minx
plt.plot([minx-delx*0.05,maxx+delx*0.05],[yeq,yeq],"b--");
```

In [144…

```
#(c)
a_1, a_2, b_1, b_2 = 3, 5/2, 2, 1

x0, y0 = 1, 5 # initial populations

timerange=6.

ts = np.linspace(0, timerange, 150)
P0 = [x0, y0] #initial conditions
Ps = odeint(dP_dt, P0, ts)
prey = Ps[:,0]
predators = Ps[:,1]

plt.figure(15)
plt.plot(ts, prey, "b-", label="Prey x")
plt.plot(ts, predators, "r-", label="Predator y")
plt.xlabel("Time")
plt.ylabel("Population")
plt.legend();
```

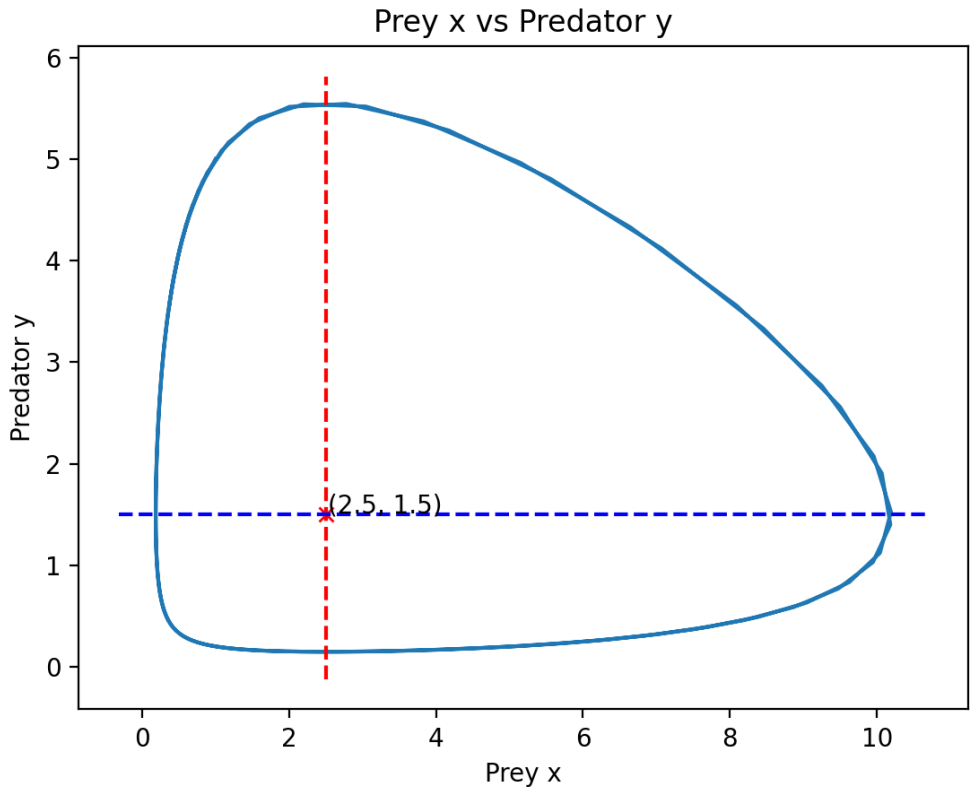


In [145…

```
plt.plot([0,timerange],[xeq,xeq],"b--")
plt.plot([0,timerange],[yeq,yeq],"r--");
```

In [146…

```
plt.figure(16)
plt.plot(prey, predators, "-")
plt.plot([xeq],[yeq],'rx')
plt.text(xeq*(1.01),yeq*(1.01),str((xeq,yeq)))
plt.xlabel("Prey x")
plt.ylabel("Predator y")
plt.title("Prey x vs Predator y");
```



In [147…

```
miny=min(predators)
maxy=max(predators)
dely=maxy-miny
plt.plot([xeq,xeq],[miny-dely*0.05,maxy+dely*0.05],"r--")

minx=min(prey)
maxx=max(prey)
delx=maxx-minx
plt.plot([minx-delx*0.05,maxx+delx*0.05],[yeq,yeq],"b--");
```

In [ ]: